



Coronary CT angiography-based FFR with ultrahigh-resolution photon-counting detector CT: Intra-individual comparison to energy-integrating detector CT

Milan Vecsey-Nagy^{a,b}, Giuseppe Tremamunno^{a,c}, U. Joseph Schoepf^a, Chiara Gnasso^{a,d}, Emese Zsarnóczyay^{a,e}, Nicola Fink^{a,f}, Dmitrij Kravchenko^{a,g}, Muhammad Taha Hagar^{a,h}, Moritz C. Halfmannⁱ, Zsófia Jokkel^b, Jim O'Doherty^{a,j}, Bálint Szilveszter^b, Pál Maurovich-Horvat^e, Pal Spruill Suranyi^a, Akos Varga-Szemes^{a,*}, Tilman Emrich^{a,i}

^a Department of Radiology and Radiological Science, Medical University of South Carolina, 25 Courtenay Dr, Charleston, SC 29425, United States

^b Heart and Vascular Centre, Semmelweis University, 68 Varosmajor Street, Budapest 1122, Hungary

^c Department of Medical Surgical Sciences and Translational Medicine, Sapienza University of Rome - Sant'Andrea University Hospital, 1035-1039 Via di Grottarossa, Rome 00189, Italy

^d Experimental Imaging Center, IRCCS San Raffaele Scientific Institute, 60 Via Olgettina, Milan 20132, Italy

^e MTA-SE Cardiovascular Imaging Research Group, Medical Imaging Center, Semmelweis University, 2 Koranyi Sandor Street, Budapest 1083, Hungary

^f Department of Radiology, University Hospital, LMU Munich, 15 Marchioninistraße, Munich 81377, Germany

^g Department of Diagnostic and Interventional Radiology, University Hospital Bonn, 1 Venusberg-Campus, Bonn 53127, Germany

^h Department of Diagnostic and Interventional Radiology, Medical Center-University of Freiburg, 55 Hugstetter Street, Freiburg im Breisgau 79106, Germany

ⁱ Department of Diagnostic and Interventional Radiology, University Medical Center of the Johannes Gutenberg-University, 1 Langenbeckstraße, Mainz 55131, Germany

^j Siemens Medical Solutions USA Inc, 40 Liberty Boulevard, Malvern, PA 19355, United States

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ABSTRACT

Purpose: To evaluate the feasibility of CT angiography-derived fractional flow reserve (CT-FFR) calculations on ultrahigh-resolution (UHR) photon-counting detector (PCD)-CT series and to intra-individually compare the results with energy-integrating (EID)-CT measurements.

Method: Prospective patients with calcified plaques detected on EID-CT between April 1st, 2023 and January 31st, 2024 were recruited for a UHR CCTA on PCD-CT within 30 days. PCD-CT was performed using the same or a lower CT dose index and an equivalent volume of contrast media. An on-site machine learning algorithm was used to obtain CT-FFR values on a per-vessel and per-patient basis. For all analyses, CT-FFR values ≤ 0.80 were deemed to be hemodynamically significant.

Results: A total of 34 patients (age: 67.3 ± 6.6 years, 7 women [20.6 %]) were included. Excellent inter-scanner agreement was noted for CT-FFR values in the per-vessel (ICC: 0.93 [0.90–0.95]) and per-patient (ICC: 0.94 [0.88–0.97]) analysis. PCD-CT-derived CT-FFR values proved to be higher compared to EID-CT values on both vessel (0.58 ± 0.23 vs. 0.55 ± 0.23 , $p < 0.001$) and patient levels (0.73 ± 0.23 vs. 0.70 ± 0.22 , $p < 0.001$). Two patients (5.9 %) with hemodynamically significant lesions on EID-CT were reclassified as non-significant on PCD-CT. All remaining participants were classified into the same category with both scanner systems.

Conclusions: While UHR CT-FFR values demonstrate excellent agreement with EID-CT measurements, PCD-CT produces higher CT-FFR values that could contribute to a reclassification of hemodynamic significance.

Abbreviations: CCTA, Coronary computed tomography angiography;; CFD, Computational fluid dynamics;; EID, Energy-integrating detector;; ICC, Intraclass correlation coefficient;; LoA, Limits of agreements;; PCD, Photon-counting detector;; TAVR, Transcatheter aortic valve replacement;; UHR, Ultrahigh-resolution.

* Corresponding author.

E-mail address: musccevi@musc.edu (A. Varga-Szemes).

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1. Introduction

While percutaneous coronary intervention has long been the cornerstone of treating obstructive coronary artery disease, studies have shown that it may be futile or even detrimental in patients without hemodynamically significant coronary stenosis [1,2]. Assessment of lesion-specific fractional flow reserve can provide insight into the functional significance of stenoses and is traditionally obtained by invasive coronary angiography [3]. With current guidelines endorsing coronary CT angiography (CCTA) over invasive angiography as the first-line modality for the evaluation of chronic coronary syndromes, the importance of reliable non-invasive alternatives to assess hemodynamic significance is underscored [4,5]. Severe clinical trials have shown that CT-FFR, based on computational fluid dynamics (CFD), can more accurately predict clinical outcomes in patients than CCTA alone and can serve as an effective risk stratification tool [6–8]. While closely correlating with CFD-derived CT-FFR values, machine learning (ML)-based CT-FFR provides additional diagnostic value over CFD calculations by optimizing computational time and cost efficiency [9].

Photon-counting detectors (PCDs) have recently been developed and introduced in CT scanners, with a dual-source system currently in clinical use [10]. Ultrahigh-resolution (UHR) mode provides a maximum in-plane resolution of 0.125 mm and a minimum slice width of 0.2 mm by directly converting absorbed X-ray photons, eliminating the need for additional septa between detector pixels [11]. The use of UHR has been shown to decrease the visualized degree of stenosis caused by calcified coronary plaques, and these differences may also alter the hemodynamic significance of lesions [12]. Recently, the feasibility of PCD-CT-based CT-FFR measurements was confirmed at standard resolution (0.6 mm), and showed a high correlation with values obtained from energy-integrating detector (EID)-CT [13]. However, there is a paucity of literature regarding CT-FFR measurements performed on UHR PCD-CT and their potential for the reclassification of hemodynamic significance in comparison to EID-CT.

The objective of this study was to assess the feasibility of performing CT-FFR calculations on UHR PCD-CT series and compare the results with EID-CT measurements in a prospective cohort of patients who underwent repeated CCTA on both scanner systems within 30 days.

2. Materials and methods

2.1. Study population

This HIPAA-compliant study was approved by the local Institutional Review Board. Written consent was obtained from participants. Consecutive patients who underwent clinical CCTA on EID-CT were prospectively enrolled between April 1st, 2023 and January 31st, 2024. The research PCD-CT scan was performed within 30 days. Inclusion criteria for the study were patients over 18 years of age undergoing CCTA for stable chest pain or planning for transcatheter aortic valve replacement (TAVR). Exclusion criteria were: contraindication to iodinated contrast media, reduced renal function (glomerular filtration rate < 45 ml/min/m²), absence of coronary calcification on EID-CT, and pregnancy. Further exclusion criteria included prior percutaneous coronary intervention, the presence of coronary artery bypass grafts, coronary anomalies, and artifacts that could compromise the accurate segmentation of coronaries. The study cohort presented in this manuscript shows partial overlap with a cohort included in a separate analysis of diagnostic accuracy for stenosis assessment and a comparative analysis of quantitative plaque volumes; both investigations are under consideration elsewhere. While all patients currently assessed have been enrolled in these unpublished works, the present study builds upon those by capturing the distinctive aspect of hemodynamic significance and includes a slightly narrowed population due to CT-FFR requirements.

2.2. CT acquisition and reconstruction

Clinical CCTAs were performed on a dual-source EID-CT (SOMATOM Force, Siemens Healthineers), following the standard-of-care protocol (collimation: 192×0.6 mm). Research CCTA scans were acquired with a dual-source PCD-CT system (NAEOTOM Alpha, software version VA50, Siemens Healthineers) in UHR mode (collimation: 120×0.2 mm). For all EID-CT scans, CARE kV and CARE Dose (Siemens Healthineers) were used to automatically determine the tube potential (90–130 kVp) and tube current (150–600 mAs) based on the patients' body habitus. Tube potential was fixed (120 kVp) for PCD-CT acquisitions, while image quality level was adjusted to closely match the expected CT dose index between scans. Patients with stable chest pain and a regular heart rate below 80 beats per minute were scanned in sequential mode (30–90 % of the R-R interval). The retrospectively gated helical scan mode was used for all pre-TAVR scans and for CCTAs in which the heart rate exceeded 80 beats per minute or arrhythmia was present. The acquisition mode and window were kept constant between the scans. For cardiac acquisitions, the same amount (56–70 ml) and type (iohexol/iopromide) of iodinated contrast material was administered for both scans at a flow rate of 3.5–4.5 ml/s.

EID-CT images were reconstructed with a slice thickness of 0.6 mm and an increment of 0.5 mm, utilizing a medium-sharp vascular kernel (Bv40), and an iterative reconstruction level of 3 (Advanced Modeled Iterative Reconstruction, Siemens Healthineers). PCD-CT datasets were reconstructed using an iterative reconstruction strength of 4 (Quantum Iterative Reconstruction, Siemens Healthineers) with a slice thickness of 0.2 mm and an increment of 0.1 mm. Kernels were selected based on BMI in a stepwise approach to ensure reasonable levels of image noise: Bv64 for BMI < 30 kg/m², Bv60 for BMI 30–40 kg/m², and Bv56 for BMI > 40 kg/m².

2.3. CT-FFR calculations

A reader with 5 years of experience in cardiovascular imaging calculated CT-FFR values using a previously described on-site ML algorithm (cFFR, version 3.5.2 prototype; Siemens Healthineers) [14,15]. Manually adjustable centerlines are first generated by the software for each coronary artery. The vessel boundaries were then calculated and manually adjusted. The proximal and distal parts of coronary stenoses were marked manually. Finally, an estimated CT-FFR value is provided for every spot on the generated mesh of the coronary tree. Measurements were performed 2 cm distal to a stenosis, at the same position on both scans (Figs. 1–3).

CT-FFR value was calculated for vessels without stenosis at the distal end of the middle arterial segment [16]. A CT-FFR value of 0.80 or less was considered hemodynamically significant [17]. For the per-patient assessment, the lowest CT-FFR value was used for analysis. A second independent reader (with 5 years of experience in cardiovascular imaging), blinded to the results of the first reader, measured CT-FFR values in 10 randomly selected participants at the same locations to evaluate reproducibility. To ensure the consistency of the location of measurements between scanners and readers, the distance of the CT-FFR values from the ostium was registered on all scans.

2.4. Statistical analysis

Statistical analysis was conducted using dedicated software (SPSS Statistics, version 29.0, IBM Corporation). Normally distributed variables are reported as mean $\pm$ standard deviation, while non-normally distributed variables were expressed as median and interquartile ranges. Categorical variables are reported as absolute frequencies and proportions. CT-FFR values were compared between the scanners using paired *t*-test. McNemar's test was used to evaluate the distribution of hemodynamically significant/non-significant vessels and patients between the two scanners. Fisher's exact test was used to compare

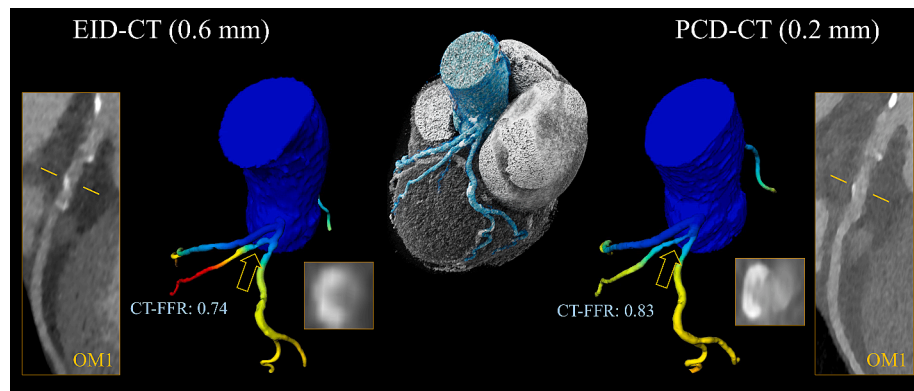


Fig. 1. Representative example of downgrade reclassification with UHR PCD-CT. Curved multiplanar reformats and cross-sectional images display a calcified plaque on the proximal segment of the first obtuse marginal branch (arrow). The resulting area stenosis appears severe on EID-CT and moderate on UHR series. The color-coded CT-FFR images show hemodynamic significance distal to the lesion on the clinical scan, but no hemodynamic significance is reached on PCD-CT. CCTA = coronary CT angiography; CT-FFR = CT angiography-derived fractional flow reserve, EID-CT = energy-integrating detector CT; OM1, first obtuse marginal = PCD-CT, photon-counting detector CT; UHR = ultrahigh-resolution.

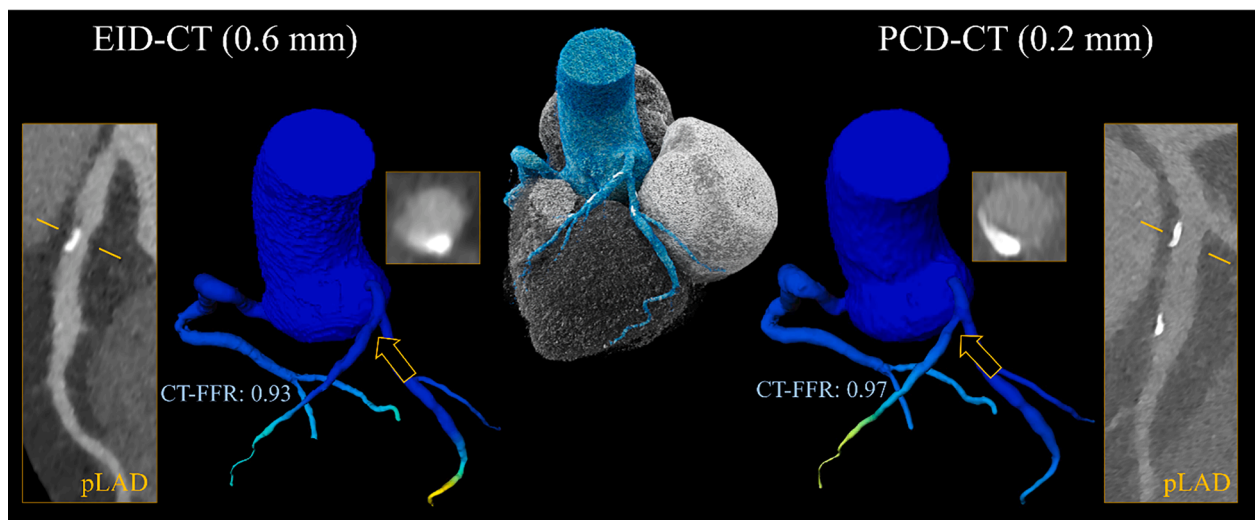


Fig. 2. Representative example for consistent grading of hemodynamic significance between scanners. A calcified plaque causes minimal stenosis in both series (arrow), as the curved multiplanar reformats and cross-sectional images demonstrate. Preserved hemodynamic properties are seen along all coronaries. CCTA = coronary CT angiography; CT-FFR = CT angiography-derived fractional flow reserve; EID-CT = energy-integrating detector CT; PCD-CT = photon-counting detector CT; pLAD = proximal left anterior descending artery; UHR = ultrahigh-resolution.

categorical variables. Pearson correlation coefficient (r) was utilized to evaluate the correlation between CT-FFR values. The agreement between CT-FFR values was assessed using two-way random-effects intraclass correlation (ICC) [18]. Mean bias and limits of agreement (LoA) are presented using Bland-Altman plots. Statistical significance was defined as two-sided p -value < 0.05 .

3. Results

3.1. Patient and acquisition parameters

A total of 320 patients were initially screened in the study. Patients were excluded for documented contrast material allergy ($n = 3$), impaired renal function ($n = 38$), and absence of visible coronary calcification on EID-CT ($n = 45$). From the 52 patients that agreed to participate, patients were excluded due to not being scanned in UHR acquisition mode ($n = 1$), extravasation of contrast material ($n = 1$), insufficient intraluminal attenuation ($n = 1$), segmentation software malfunction due to situs inversus ($n = 1$), prior percutaneous coronary intervention ($n = 5$), implanted coronary artery bypass graft ($n = 2$), and

marked misalignment artifacts ($n = 7$). The final cohort included 34 eligible patients (mean age, 67.3 ± 6.6 ; 7 women [20.6 %]). Stable chest pain was the indication for 29 patients, and work-up prior to planned TAVR for 5 participants (Table 1).

A median interval between scans was 14.0 (7.0–19.8) days. Mean heart rate did not differ significantly between EID-CT and PCD-CT scans (69.4 ± 8.5 vs. 67.4 ± 9.4 beats per minute, respectively, $p = 0.47$). A median of 70.0 (56.0–70.0) ml of contrast media was used for the scans. Effective dose was significantly lower for PCD-CT scans (12.5 [IQR, 10.5–14.2] vs. 15.4 [IQR, 11.6–20.2] mSv, $p < 0.001$), as summarized in Table 2.

3.2. Comparison of CT-FFR values obtained from the two different scanners

A total of 102 coronary arteries were analyzed. PCD-CT-derived CT-FFR values were found to be significantly higher than EID-CT values at both the vessel (0.58 ± 0.23 vs. 0.55 ± 0.23 , $p < 0.001$) and patient levels (0.73 ± 0.23 vs. 0.70 ± 0.22 , $p < 0.001$). There was a strong, significant correlation between CT-FFR values obtained from the

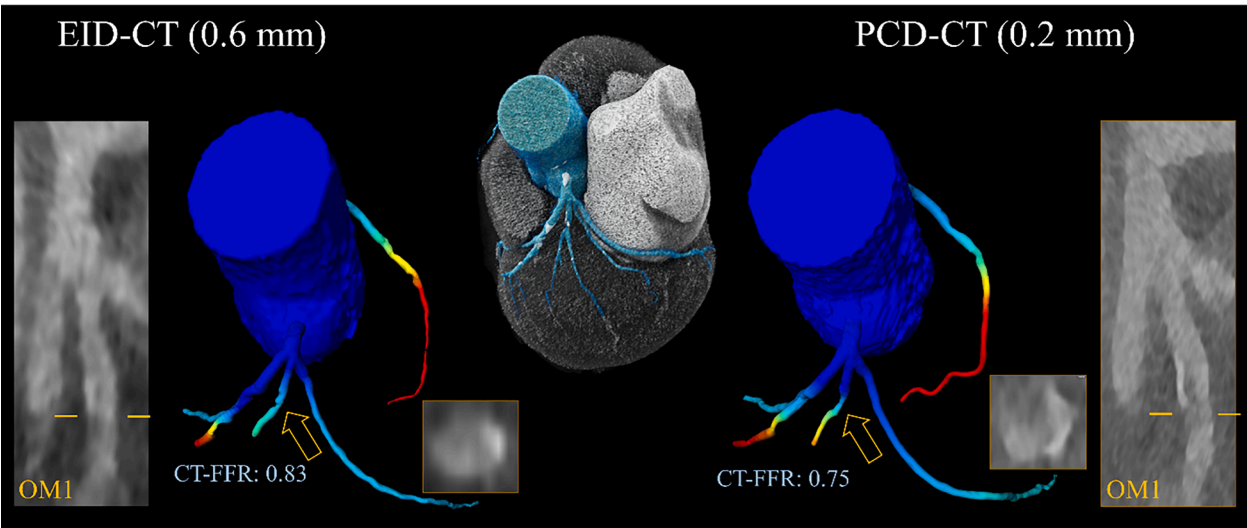


Fig. 3. Upward reclassification of hemodynamic significance with PCD-CT. The color-coded CT-FFR images demonstrate hemodynamic significance distal to a partially calcified plaque on the obtuse marginal branch on PCD-CT (arrow). However, no hemodynamic significance is reached on EID-CT. A patient-based assessment revealed consistent results between the two scanners, with both identifying hemodynamically significant stenoses along the left anterior descending and right coronary arteries. CCTA = coronary CT angiography; CT-FFR = CT angiography-derived fractional flow reserve; EID-CT = energy-integrating detector CT; OM1 = first obtuse marginal; PCD-CT = photon-counting detector CT; UHR = ultrahigh-resolution.

Table 1
Descriptive patient data.

Total (n = 34)	
Age (years)	67.3 ± 6.6
Female sex, n (%)	7 (20.6)
BMI (kg/m ²)	31.3 ± 4.3
Race	
Caucasian, n (%)	30 (88.2)
African American, n (%)	3 (8.8)
Native American, n (%)	1 (3.0)
Cardiovascular risk factors	
Hypertension, n (%)	27 (79.4)
Diabetes, n (%)	8 (23.5)
Dyslipidemia, n (%)	24 (70.6)
Current smoking, n (%)	2 (5.9)
Positive family history, n (%)	7 (20.6)
Indication	
Stable chest pain, n (%)	29 (85.3)
TAVR planning, n (%)	5 (14.7)
Coronary artery calcium score	442.0 (173.0–503.0)

Continuous data are presented as mean ± standard deviation or median and interquartile ranges, while categorical variables are shown as numbers and percentages.

BMI = body mass index; TAVR = transcatheter aortic valve replacement.

scanners in both per-vessel ($r = 0.87$, $p < 0.001$) and per-patient ($r = 0.89$, $p < 0.001$) analyses. An excellent agreement was shown between CT-FFR values obtained from the scanners for the vessels (ICC: 0.93 [0.90–0.95]) and on a per-patient basis (ICC: 0.94 [0.88–0.97]). Bland Altman analysis demonstrated similar mean bias and LoA ranges for the vessels (mean bias, 0.03; LoA, −0.19/0.25) as for the patients (mean bias, 0.03; LoA, −0.18/0.25). Inter-reader agreement was excellent for CT-FFR measurements (ICC: 0.90 [0.83–0.94]). Fig. 4 demonstrates the scatterplot and Bland-Altman analysis of CT-FFR values and Table 3 summarizes the comparison between EID-CT and PCD-CT.

3.3. Distribution of hemodynamically significant lesions

A high proportion of vessels contained hemodynamically significant stenoses on EID-CT (60/102, 58.8 %) and PCD-CT (50/102, 49.0 %). A total of 11 vessels characterized as hemodynamically significant on EID-CT were considered non-significant on PCD-CT, and one vessel was

Table 2
CT acquisition and reconstruction parameters.

	EID-CT	PCD-CT	p
Heart rate (beats/min)	69.4 ± 8.5	67.4 ± 9.4	0.468
Scan mode			–
Sequential, n (%)	27 (79.4)		
Spiral, n (%)	7 (20.6)		
Tube potential (kVp)			<0.001
90, n (%)	4 (11.8)	–	
110, n (%)	13 (38.2)	–	
120, n (%)	11 (32.4)	34 (100.0)	
130, n (%)	6 (17.6)	–	
Tube current (mAs)	254.0	–	
	(229.0–276.0)		
Image quality level	–	64.0	
		(64.0–80.0)	
Amount of contrast material (ml)	70.0		–
	(56.0–70.0)		
Acquisition time (s)	6.7	9.0	<0.001
	(5.0–6.7)	(9.0–9.0)	
CTDI _{vol} (mGy)	43.3	37.3	<0.001
	(33.2–51.8)	(32.9–42.6)	
Dose length product (mGy*cm)	629.0	520.0	<0.001
	(475.0–817.0)	(437.5–592.5)	
Effective dose (mSv)	15.4	12.5	<0.001
	(11.6–20.2)	(10.5–14.2)	
Slice thickness (mm)	0.6	0.2	
Slice increment (mm)	0.5	0.2	
Kernel			<0.001
Bv64, n (%)	–	16 (47.0)	
Bv60, n (%)	–	17 (50.0)	
Bv56, n (%)	–	1 (3.0)	
Bv40, n (%)	48 (100.0)	–	
Iterative reconstruction strength	3 (ADMIRE)	4 (QIR)	

Continuous data are presented as mean ± standard deviation or median and interquartile ranges, while categorical variables are shown as numbers and percentages.

Continuous parameters were compared with paired sample *t*-test or Wilcoxon test, while comparison of categorical variables was performed using Fisher's exact test.

The effective radiation dose of CT was calculated by multiplying the dose-length product by a cardiac-specific conversion factor of 0.026 mSv·mGy^{−1} cm^{−1}. ADMIRE = Advanced Modeled Iterative Reconstruction; CTDI_{vol} = volume computed CT dose index; EID-CT = energy-integrating detector CT; PCD-CT = photon-counting detector CT; QIR = Quantum Iterative Reconstruction.

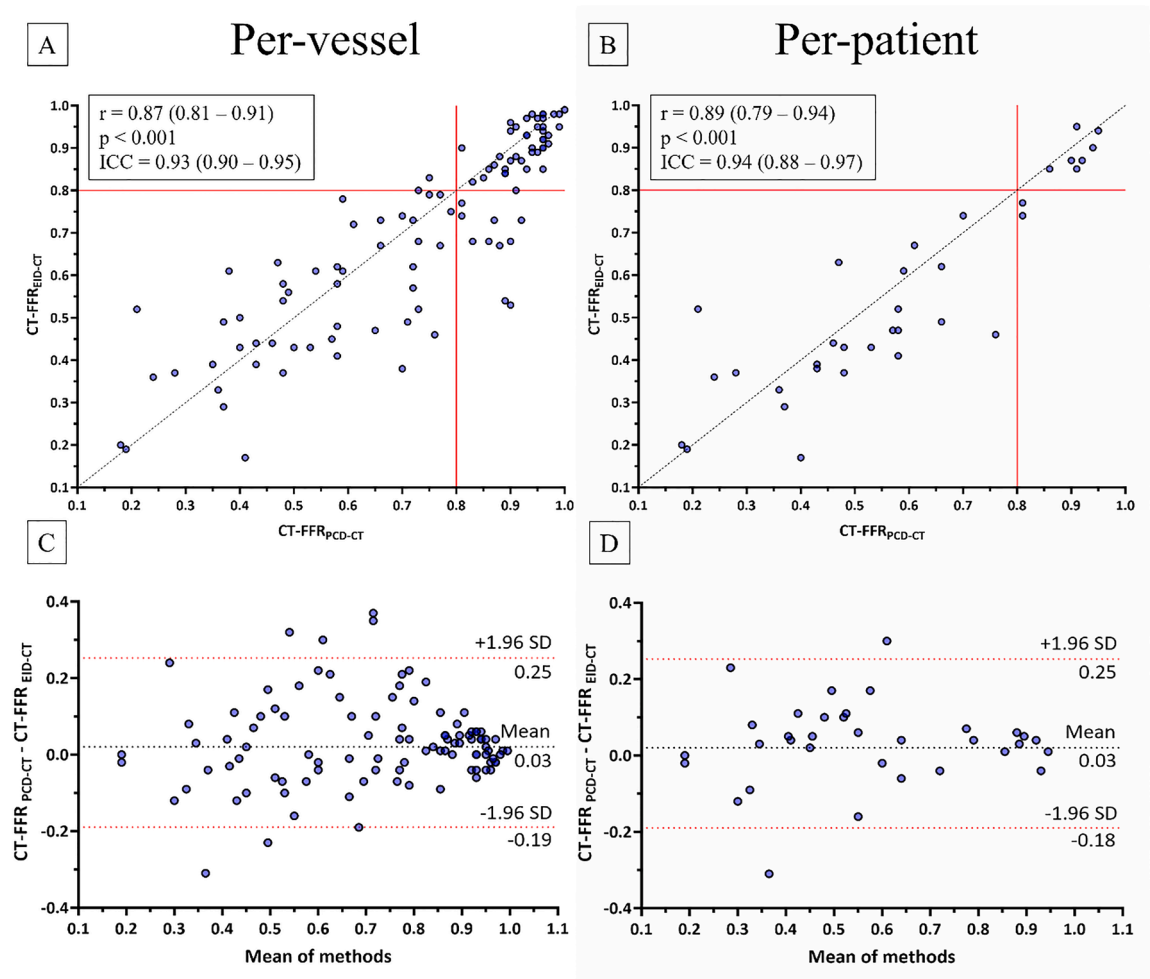


Fig. 4. Per-vessel and per-patient comparisons between EID-CT and PCD-CT. A strong correlation and excellent agreement are seen between scanner systems for vessels (A) and on a patient basis (B). Per-vessel (C) and per-patient (D) Bland-Altman analysis demonstrate no clear systematic bias and similar limits of agreement (red dotted lines) for both comparisons. CT-FFR = CT angiography-derived fractional flow reserve; EID-CT = energy-integrating detector CT; ICC = intraclass correlation coefficient; PCD-CT = photon-counting detector CT.

Table 3
Comparison of CT-FFR values obtained on EID-CT and PCD-CT.

	CT-FFR _{EID-CT}	CT-FFR _{PCD-CT}	p	r	ICC (95 % CI)	Bias	LoA
Per-vessel (n = 102)	0.70 ± 0.22	0.73 ± 0.23	<0.001	0.87	0.93 (0.90–0.95)	0.03	−0.19/0.25
Per-patient (n = 34)	0.55 ± 0.23	0.58 ± 0.23	<0.001	0.89	0.94 (0.88–0.97)	0.03	−0.18/0.25

Continuous data are presented as mean ± standard deviation.
CT-FFR were compared using paired samples *t*-test.
CT-FFR = CT angiography-derived fractional flow reserve; EID-CT = energy-integrating detector CT; ICC = intraclass correlation coefficient; LoA = limits of agreement; PCD-CT = photon-counting detector CT.

considered hemodynamically significant only on PCD-CT. There were no significant differences observed between EID-CT and PCD-CT in terms of the distribution of hemodynamically significant stenoses for the left anterior descending artery (24/34, 70.6 % vs. 21/34, 61.8 %, $p = 0.25$), the left circumflex artery (19/34, 55.9 % vs. 14/34, 41.2 %, $p = 0.13$), and the right coronary artery (17/34, 50.0 % vs. 15/34, 44.1 %, $p = 0.25$). Overall, 2 patients were assigned to the CT-FFR positive group by EID-CT and reclassified to the hemodynamically non-significant group by PCD-CT. The remaining 32 patients were consistently assigned using both scanner systems. On a patient level, there were no significant differences in the distribution of hemodynamically significant/non-significant characterization (27/34, 79.4 % vs. 25/34, 73.5 %, $p = 0.50$). Table 4 shows the distribution of hemodynamically significant

lesions on a per-vessel and per-patient basis.

4. Discussion

The use of UHR has been demonstrated to reduce the visualized degree of coronary stenosis compared to standard resolution, and these differences may also influence the hemodynamic significance of lesions. The current study aimed to assess the feasibility of obtaining CT-FFR measurements from UHR PCD-CT datasets and provide an intra-individual comparison to conventional EID-CT. The major findings are: (1) CT-FFR values obtained from UHR series demonstrated a strong correlation and excellent agreement with EID-CT measurements. (2) PCD-CT produced higher CT-FFR values that could lead to a

Table 4
Distribution of hemodynamically significant lesions on a per-vessel and per-patient basis.

	CT-FFR _{EID} CT ≤ 0.80	CT-FFR _{EID} CT > 0.80	CT-FFR _{PCD} CT ≤ 0.80	CT-FFR _{PCD} CT > 0.80	p
Per-vessel					
LAD	24 (70.6)	10 (29.4)	21 (61.8)	13 (38.2)	0.246
LCX	19 (55.9)	15 (44.1)	14 (41.2)	20 (58.8)	0.128
RCA	17 (50.0)	17 (50.0)	15 (44.1)	19 (55.9)	0.498
Per-patient	27 (79.4)	7 (20.6)	25 (73.5)	9 (26.5)	0.496

Values are presented as n (%).
Differences in the distribution are tested with McNemar's test.
CT-FFR = CT angiography-derived fractional flow reserve; EID-CT = energy-integrating detector CT; LAD = left anterior descending artery; LCX = left circumflex artery; PCD-CT = photon-counting detector CT; RCA = right coronary artery.

reclassification of hemodynamic significance on both vessel and patient levels.

UHR PCD-CT has been shown to improve the accuracy of calcified stenosis assessment and the evaluation of stents, as well as resulting in a clinically relevant rates of Coronary Artery Disease Reporting and Data System reclassification compared to EID-CT [19–23]. While the growing clinical evidence supporting UHR CCTA indicates that the number of patients undergoing PCD-CT is expected to expand, CT-FFR models were originally developed for EID-CT, and there is limited information available regarding the feasibility and performance of on-site CT-FFR calculations on PCD-CT datasets. The accuracy of CFD-based CT-FFR measurements for differentiating functionally significant coronary disease has been demonstrated in multicenter trials [24–26]. Although several alternative on-site CT-FFR algorithms have been developed with the goal of simulating CFD-based measurements, few have been validated in large-scale investigations [27–29]. The current algorithm has been shown to perform at a level comparable to CFD-based calculations, without the need for extensive computational time [9,30].

To date, only one study has provided an intra-individual comparison of CT-FFR between EID-CT and PCD-CT. Zsarnoczay et al. demonstrated that obtaining CT-FFR measurements from PCD-CT is feasible using standard resolution (0.6 mm), with good correlation to EID-CT [13]. The current investigation extends upon this previous study by demonstrating the feasibility of UHR-based CT-FFR measurements, despite the increased complexity of UHR series. While UHR PCD-CT-derived CT-FFR also exhibited excellent agreement with the EID-CT measurements, it manifested in significantly higher CT-FFR values. The findings may be attributed to the enhanced delineation of calcified lesions and the subsequent reduction in stenosis, which could have implications for techniques that rely on accurate lumen visualization, such as CT-FFR. Previous investigations have demonstrated that the ML-based algorithm underestimates FFR values in comparison to the invasive reference [9,14]. However, further large-scale validation studies are necessary to confirm whether the currently documented differences translate to higher accuracy with UHR PCD-CT.

Despite significant differences in CT-FFR measurements, only a relatively low proportion of patients were reclassified to hemodynamically non-significant with UHR. This may be partly attributed to the fact that a substantial proportion of patients were CT-FFR positive due to extensive coronary calcification. The patients included in the study are representative of those who would undergo clinical UHR imaging due to advanced calcification seen on non-enhanced scans. Consequently, although partial volume averaging contributes to the diagnostic ambiguity of CT-FFR in clinical practice, patients were not excluded based on extensive calcification to simulate the feasibility of performing UHR CT-FFR measurements in a real-world setting [31]. The current study is distinguished from previous investigations in that it does not include

highly selected patient groups or specific types of lesions [9,32,33].
The present study had limitations that merit consideration. First, the study population was limited to 34 patients. Second, the readers may have been able to distinguish between the PCD-CT and the EID-CT scans based on the appearance of the series, which may have introduced a bias. Third, the differences in tube potential, the slightly lower radiation dose for PCD-CT, and the different reconstruction settings for the scanner systems may have influenced the results. Fourth, a relatively high proportion of patients were excluded due to severe misalignment artifacts in at least one of the scans, warranting the need for novel post-processing algorithms that may reduce these artifacts and make CT-FFR accessible to a wider range of patients [34]. Finally, it is currently unknown whether the differences in CT-FFR values translate to higher accuracy, as EID-CT and PCD-CT measurements were not validated against invasive FFR.

5. Conclusions

The use of UHR PCD-CT results in significantly higher CT-FFR values when compared to dual-source EID-CT in a within-patient design. The strong correlation and agreement between CT-FFR values obtained from different scanner systems, and the low proportion of reclassification in terms of hemodynamic significance when UHR is used, suggest that UHR may lead to less pronounced clinical implications in the intended target population, as opposed to mere stenosis degree reclassification. The question of whether UHR PCD-CT offers a more clinically relevant reclassification rate in a population with less severe coronary artery disease and a lower proportion of patients in the hemodynamically significant category should be further investigated.

Declaration of conflicting interest

Dr. Schoepf receives institutional research support and/or honoraria for speaking and consulting from Bayer, Bracco, Elucid BioImaging, Guerbet, HeartFlow Inc., Keya Medical, and Siemens Healthineers. Dr. Szilveszter and Dr. Maurovich-Horvat receive institutional research and travel support from Siemens Healthineers. Dr. Varga-Szemes receives institutional research and travel support from Siemens Healthineers and is a consultant and shareholder for Elucid Bioimaging. Dr. Emrich is a consultant for Circle Cardiovascular Imaging Inc. and receives travel support, speaker fee and institutional research support from Siemens Healthineers.

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CRediT authorship contribution statement

Milan Vecsey-Nagy: Conceptualization, Data curation, Methodology, Writing – original draft. **Giuseppe Tremamunno:** Investigation. **U. Joseph Schoepf:** Resources. **Chiara Gnasso:** . **Emese Zsarnóczay:** Investigation. **Nicola Fink:** Investigation. **Dmitrij Kravchenko:** Writing – review & editing. **Muhammad Taha Hagar:** Data curation. **Moritz C. Halfmann:** Writing – review & editing. **Zsófia Jokkel:** Data curation. **Jim O'Doherty:** Supervision, Writing – review & editing. **Bálint Szilveszter:** Data curation, Writing – review & editing. **Pal Maurovich-Horvat:** Writing – review & editing. **Pal Spruill Suranyi:** Writing – review & editing. **Akos Varga-Szemes:** Methodology, Supervision, Writing – review & editing. **Tilman Emrich:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

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